ECO ASSISTANT QUICK REFERENCE

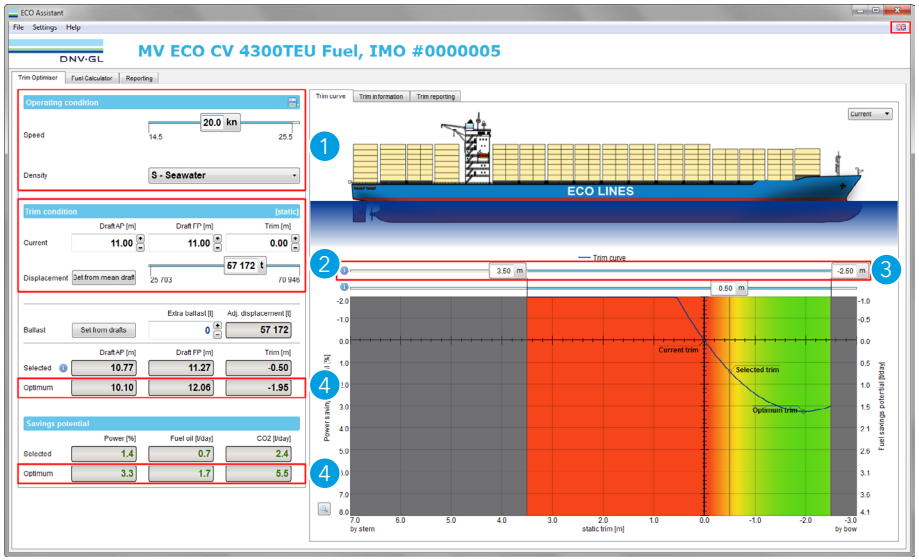
Trim Optimiser

How to ensure optimal trim

Standard procedure

Ensure that the dongle is placed in the PC before starting ECO Assistant.
Choose preferred language by clicking on the 'flag'-symbol (0)

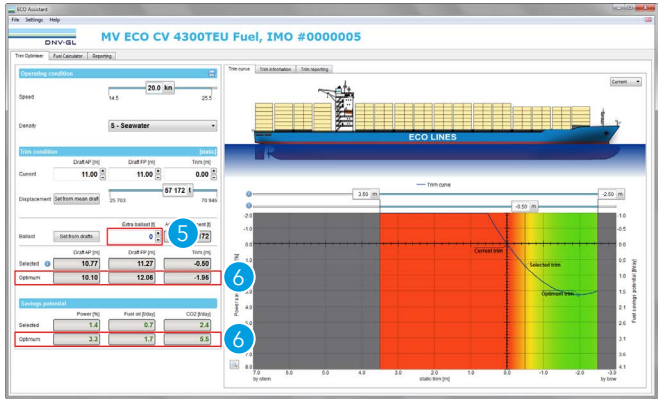
- Select target (water) speed and appropriate water to sail in for the voyage leg ahead (1)
- Enter current departure sailing condition (direct readings from draft marks or from loading computer) by setting draft AP and FP or mean draft and trim (2)
- Adjust the trim range limiter according to the permitted limits from stability book, rules, etc. (3)
- ECO Assistant indicates the optimum trim (assuming no additional ballast water needed) and the approximate daily fuel/CO_2 savings (4)



Tool is to inform crew of hydrodynamically optimal trim – it does not assess implications on stability, bending moments and the like. The sole responsibility for the ship continues to rest with the crew.

If extra ballast water is needed to achieve recommended trim …

- Enter additional ballast water, if this is required to achieve the recommended trim. For less ballast water use negative quantity (5)
- Tool indicates fuel/CO_2 savings with optimised trim (solid curve) considering additional ballast water (6)



- If the savings from this trim are off-set by additional displacement, the digits will turn red: then re-trimming does not make sense
- If extra ballast water has a positive effect, it makes sense to take more ballast as required!

Set selected trim condition for voyage ahead

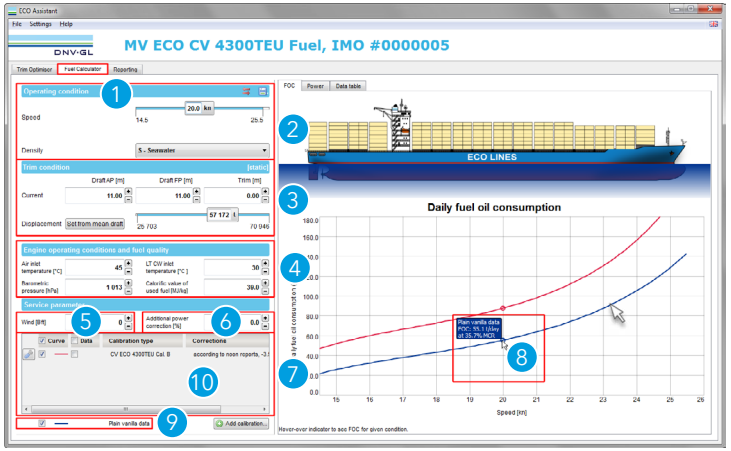
- Set selected trim condition with the help of the slider (7)
- ECO Assistant indicates fuel/CO_2 savings for new trim (solid curve) considering additional ballast water (8)

Fuel Calculator

How to determine daily FOC

Standard procedure

- Click on the 'Fuel Calculator' tab (1)
- Select speed of interest and appropriate water to sail in (2)
- Enter sailing condition of interest (3)
- Adjust engine operating conditions and fuel quality (4)
- Select wind speed (5) based on Beaufort scale. Please note that this setting includes the influence of wind correlated waves, if required an additional power correction can be applied (6). Both parameters affect all calibrated curves.
- Choose the fuel consumption curves to be shown by tagging the appropriate curve check boxes within the calibration window (7)
- Selected fuel consumption curves are displayed in the diagram on the right side. Hovering over the speed indicator shows the absolute fuel oil consumption for the selected curve (8)



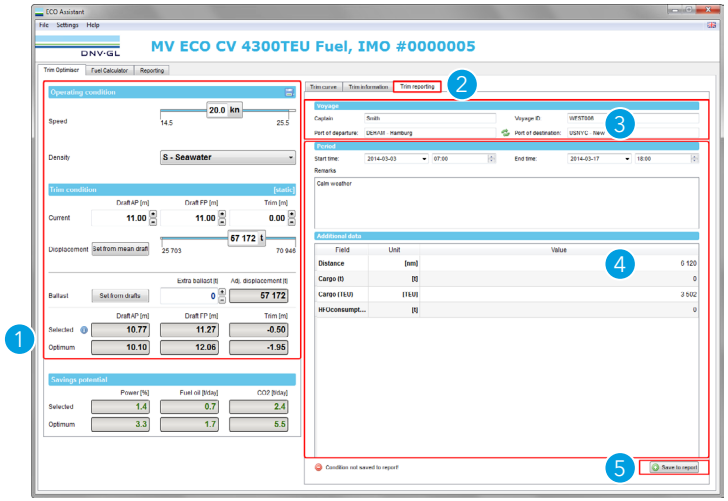
This tool is to inform crew of fuel oil consumption for specified operating conditions. Commonly plain vanilla data (9) and baseline (10) should be available for all ships. Individualised calibration files based on noon-to-noon report analysis must be ordered separately.

Reporting

How to document trim optimisation results

Save trim condition to report

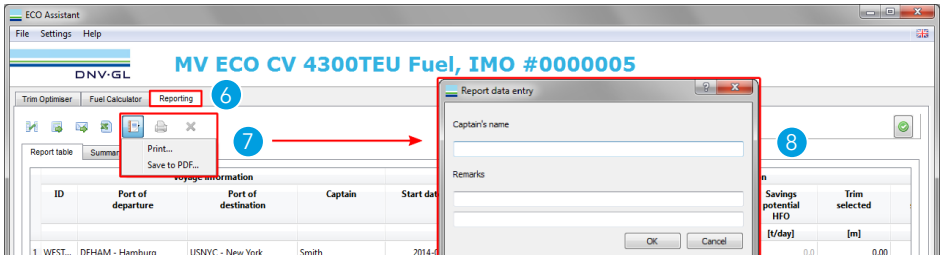
- Click on the trim optimiser tab and check that the entered values and the selected sailing trim shall be reported (1)



- Activate the 'Trim reporting' tab within the Trim Optimiser (2)
- Add report data by entering basic voyage data (3) and specific data for the period the selected condition is sailed (4)
- Click on button 'Save to report' (5)

Generate trim report for the filtered data

- Within Reporting (6) click on the 'Trim report' button (7)
- Choose between print and save to PDF format
- Enter captain's name and remarks to dialog box and click 'OK' (8)



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